Problem 00000001363: Thompson's V is not locally finite

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Result and scope. The English source asserts that, among locally finite groups meeting its action requirement, the minimum group is Thompson's V. This necessarily requires V itself to be locally finite. We refute that necessary membership using the actual Cantor-space model of V: two genuine prefix-transposition homeomorphisms have a product of infinite order, and its actual one-generator subgroup is infinite.

Here *locally finite* has its ordinary group-theoretic meaning: every subgroup generated by a finite subset is finite. We do not define a fictitious E_∞ action or an optimization degree, prove an alternative minimal group, or claim to formalize descriptive-set-theoretic universality. The Chinese wording has an undefined ℓ qualification, and neither language specifies a comparison degree; no conclusion about an undefined alternative property is asserted. The counterexample concerns the explicit ordinary locally finite qualification in the English necessary condition.

1. The actual standard group model

Use Cantor space $\mathcal{C} = \{0,1\}^{\mathbb{N}}$ with its genuine product topology, finite words $u \in \{0,1\}^*$, and actual concatenation us for $s \in \mathcal{C}$. Its cone is the actual set $u\mathcal{C} = \{us : s \in \mathcal{C}\}$. For disjoint cones $u\mathcal{C}, v\mathcal{C}$, a prefix transposition is the homeomorphism satisfying all three rules

$$t(us) = vs, \qquad t(vs) = us, \qquad t(x) = x \quad (x \notin u\mathcal{C} \cup v\mathcal{C}).$$

These full laws, including fixedness outside the two cones, are the actual Lean predicate `PrefixSwap`. Every such map is proved involutive. The formal group `thompsonV` is the subgroup of $\mathrm{Homeo}(\mathcal{C})$ generated by *all* such prefix transpositions, not just the two maps used below.

This is the standard transposition-generated model in Bleak–Quick [1], printed p. 3 for Cantor space and p. 6, §2, equation (2.1), for the complete transposition laws and finite-product model. Belk et al. [2], printed pp. 420–421, also define V by actual finite prefix substitutions. We adopt this standard concrete group model; we do not claim to reprove equivalence with every abstract finite presentation or every alternate realization of V.

2. Two real continuous involutions

Lean constructs two functions on infinite binary sequences with genuine inverses, proves both directions continuous in the product topology, and packages them as actual `Homeomorph`s:

a	$0s \mapsto 1s$	$1s \mapsto 0s$	
b	$00s \mapsto 1s$	$1s \mapsto 00s$	$(01s \mapsto 01s)$.

The proof verifies $a^2 = b^2 = 1$ on every infinite sequence. For b, the three cases $00s$, $01s$, and $1s$ exhaust the space. Continuity follows from actual coordinate projections, continuous prepend/tail maps, and the clopen coordinate conditions used in the piecewise definitions. No continuity or inverse field is an assumption.

The actual cones $0\mathcal{C}$ and $1\mathcal{C}$ are disjoint, as are $00\mathcal{C}$ and $1\mathcal{C}$, by evaluating the first coordinate. Both maps satisfy the full corresponding prefix-swap laws, so they belong to the standard full group V by its genuine subgroup closure.

3. Infinite order from an actual infinite orbit

Let $f = b \circ a$, using the usual function-composition convention. Since both factors are in V, so is f. The essential identity, proved for every tail s, is

$$f(0s) = 00s.$$

Define the actual sequence $x_n \in \mathcal{C}$ by

$$x_n(i) = \begin{cases} 0, & i < n+1, \\ 1, & i \geq n+1. \end{cases}$$

Lean proves $f(x_n) = x_{n+1}$ and then $(f^n)(x_0) = x_n$ for every natural n. If $n > 0$, the sequence x_n has coordinate 1 equal to 0, whereas $x_0(1) = 1$. Thus $f^n \neq 1$ for every positive n, so f has genuine infinite order. No orbit length or element order is assigned from an expected answer.

Now regard f as the actual subtype element `witness` of V, and form the actual subgroup

$$H = \langle f \rangle \leq V$$

using `Subgroup.closure` of its singleton. The named theorem `cyclicWitness_single_generator` gives a real generating finite set of cardinality one. The named theorem `cyclicWitness_infinite` proves its actual carrier is an infinite set. Indeed, if H were finite, its actual element f would have finite order by the finite-group theorem; the subgroup inclusions would give a positive power $f^n = 1$ in $\mathrm{Homeo}(\mathcal{C})$, contradicting the orbit calculation.

4. The necessary minimum-membership condition fails

The formal locally finite predicate is stated directly on real groups:

$$\forall S \subseteq \Gamma \text{ finite}, \quad \langle S \rangle \text{ is finite.}$$

The same true subgroup-closure argument proves every locally finite group is torsion. Applying it to the concrete $f \in V$, or applying the explicit infinite H directly, gives

$$\neg\,\mathrm{LocallyFinite}(V).$$

For every additional property P of the actual subgroup, Lean also proves

$$\neg\big(\mathrm{LocallyFinite}(V) \ \wedge \ P(V)\big).$$

This universal statement does not pretend that P defines a Borel action: it records that any further action or optimization requirement cannot restore the already false local-finiteness qualification. A minimum among qualified candidates must first be a qualified candidate. Therefore the source's stated minimum cannot be this V under the ordinary locally finite reading, independently of the unspecified degree or any additional action requirement.

5. Verification, attribution and limits

All 49 named declarations, including 32 theorems and proof-bearing homeomorphism definitions, are axiom-audited. The package contains a new pinned root build, warning-as-error audit and full fresh kernel replay with source-bound receipts. The original bilingual source, full LaTeX and this PDF are preserved. No `sorry`, `native_decide` or extra axiom is used. Both pages were rendered and visually checked by worker and separate supervising AI. No earlier same-ID submission was found in the recorded repository all-state search.

The infinite-order obstruction is elementary; its actual topology, full-group membership and generated-subgroup bridges are the formal contribution. Work was led by a Codex worker with separate AI supervision of source fidelity and document layout. Organizer acceptance and human verification are not asserted.

References

[1] C. Bleak and M. Quick, *The infinite simple group V of Richard J. Thompson: presentations by permutations*, fixed arXiv:1511.02123v2, pp. 3 and 6, §2, (2.1).

[2] J. Belk, C. Bleak, M. Quick and R. Skipper, *Type systems and maximal subgroups of Thompson's V*, Trans. Amer. Math. Soc. Ser. B 12 (2025), 417–469, author-accessible published PDF, pp. 420–421, Definition 1.1.